INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II ,Third Semester Examination, October- November 2026

Subject Code: BCAAE301

Subject Name: Environmental Studies

Set No:1

Full Marks: 30 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. Define plume and stack. **2**
2. Arrange the steps of noise pollution control. **2**
3. Give examples of four ecosystem components. **2**
4. What is rooftop rainwater harvesting? **2**
5. Explain the factors on which the BOD rate constant depends. **2**
6. State any two objectives of Environmental Impact Assessment. **2**

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Explain nitrogen fixation and nitrification processes. **5**
8. Describe the harmful effects of improper disposal of electronic waste on human health and the environment. **5**
9. A **40 mL** wastewater sample is diluted to **300 mL**. The DO initially is **8.9 mg/L** and after 5 days is **3.5 mg/L**. Calculate the BOD_5. **5**

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Describe the carbon cycle in the environment with a sketch diagram. **10**
11. What is e-waste? Explain the major sources, harmful effects, and proper methods of e-waste management. **10**

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Full Marks: 30 Set No:2 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. What is green computing? Give two examples of how it is practiced. **2**
2. Represent sound intensity in decibel. **2**
3. What are producers and consumers in ecosystem? **2**
4. State the classification of soil. **2**
5. What is mixing depth? **2**
6. Outline two importance of hydrological cycle. **2**

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Describe the role of technology in making cities cleaner, safer, and more sustainable. **5**
8. Discuss briefly on food web occurring in nature. **5**
9. A **300 mL bottle** is filled with **30 mL wastewater** and the rest with unseeded dilution water. The initial DO is **8.5 mg/L** and after 5 days it is **3.5 mg/L**. Calculate the BOD_5. **5**

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Describe the nature of pyramids of energy taking place in the food chain with explanations. **10**
11. Discuss oxygen cycling occurring between different environmental segments. **10**

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Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. Arrange the steps of noise pollution control. **2**

2. What is the nature of energy pyramids and why? **2**

3. State any two consequences of global warming. **2**

4. How soil is formed? **2**

5. Why inner core of the earth is solid instead of higher temperature than the outer core? **2**

6. Define atmosphere, hydrosphere, lithosphere and biosphere. **2**

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Explain the nature of parasitic food chain with diagram. **5**

8. A **300 mL bottle** is filled with **10 mL wastewater** and the rest with unseeded dilution water. The initial DO is **8.6 mg/L** and after 5 days it falls to **5.6 mg/L**. Calculate the BOD_5. **5**

9. Discuss the major causes of climate change and explain how climate change can affect human settlements. **5**

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Describe any four major environmental movements in India, such as the Chipko Movement, Appiko Movement, Silent Valley Movement, and Narmada Bachao Andolan. Explain their objectives and environmental significance. **10**

11. Discuss on nitrogen cycling occurring between different environmental segments. **10**

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